

Asking Testable Analytical Question

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HES 3000-002, Sport Performance Analytics

Start With Something You Notice

Think of one observation, concern, or belief about sport that you would like to investigate.

Examples:

- “Our team fades late in games.”
- “This prospect looks better than the statistics suggest.”
- “Some athletes do not recover well after hard practices.”

Write yours in one sentence. Does not need to be a formal question yet.

Analysis must start with clarity

“Our team fades late in games” leaves several things unresolved:

- What does **fades** mean?
- Which team, players, games, or situations?
- Compared with what?
- What evidence would support or challenge the belief?
- What decision would change if the belief were true?

The starting point must be observable and the purpose must be clear.

Good Questions Connect Curiosity to a Decision

A useful analytical question identifies:

1. **The decision or purpose** : Why do we need the answer?
2. **The outcome** : What are we trying to understand?
3. **The comparison** : Relative to whom, what, or when?
4. **The context** : Under which conditions?
5. **The evidence** : What observations could answer it?

The question is the bridge between the sport problem and the analysis.

Begin With the Decision

“I want to know whether our players are tired.”

Possible decisions include:

- Should today's practice be modified?
- Should a player's minutes be restricted tomorrow?
- Should the weekly training plan change?
- Should the team adopt a new recovery strategy?

The same concern can lead to different questions because the decisions are different.

Define the Outcome Before Choosing the Metric

“tired” could refer to *fatigue*, which could mean:

- Reduced physical performance
- Increased perceived effort
- Muscle soreness
- Lower motivation or alertness
- Incomplete recovery from recent training

A metric represents one part of a concept; it is not automatically the concept itself.

Comparisons Give Numbers Meaning

Consider: “Did the athlete complete a high workload?”

High compared with:

- The athlete's usual workload?
- The planned workload?
- Teammates at the same position?
- Competition demands?
- A threshold associated with a particular goal?

Without a reference point, “high,” “low,” “good,” and “poor” are usually empty labels.

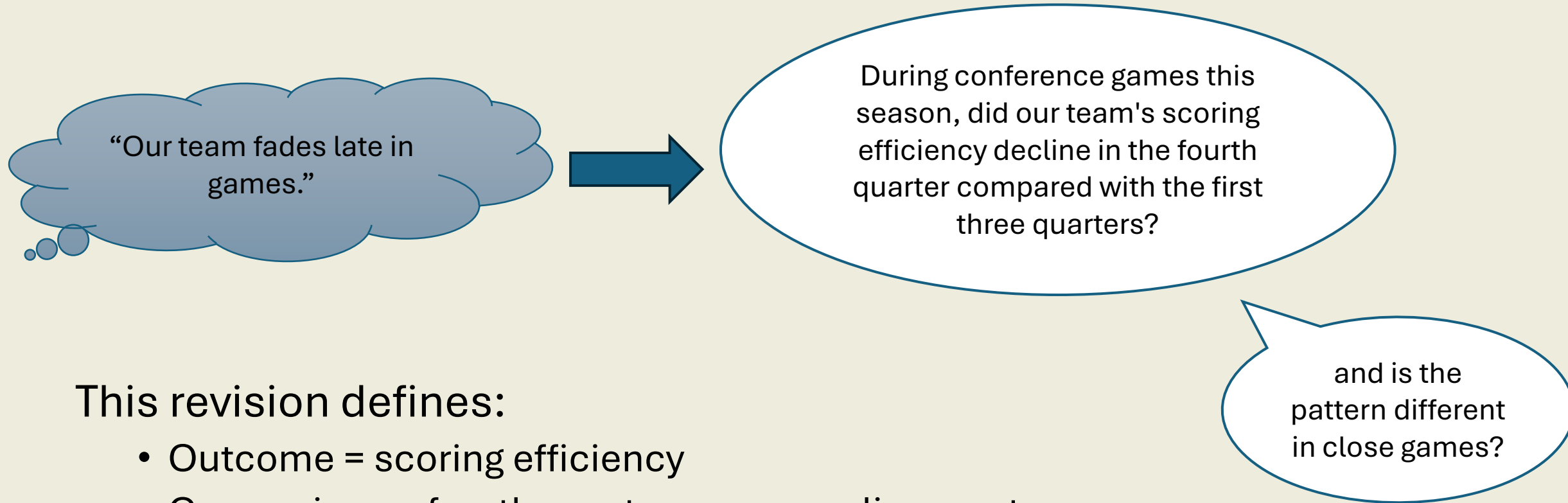
Context Determines What the Comparison Means

The same performance value may mean something different depending on:

- Position or role
- Opponent quality
- Game state
- Training phase
- Injury or return-to-play status
- Age and competition level

Useful questions define just enough context to make the answer interpretable.

From Concern to Question

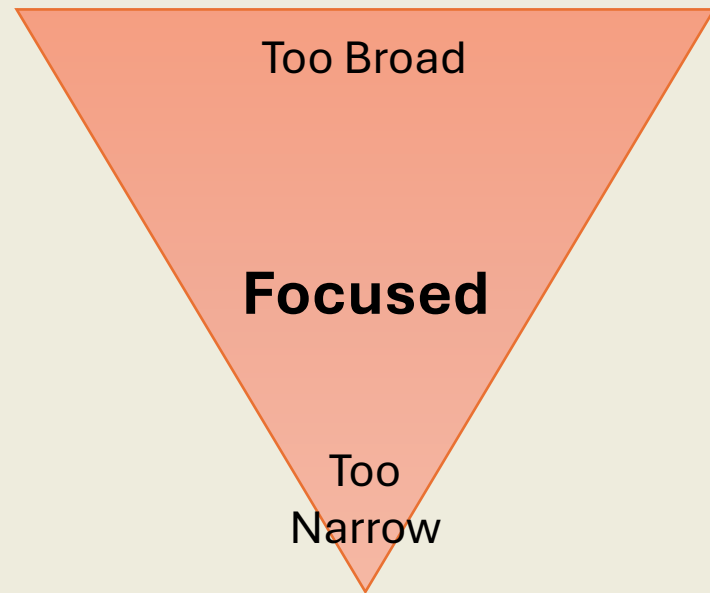


This revision defines:

- Outcome = scoring efficiency
- Comparison = fourth quarter versus earlier quarters
- Context = conference games

What else needs to be resolved?

Precision Does Not Mean Small Questions

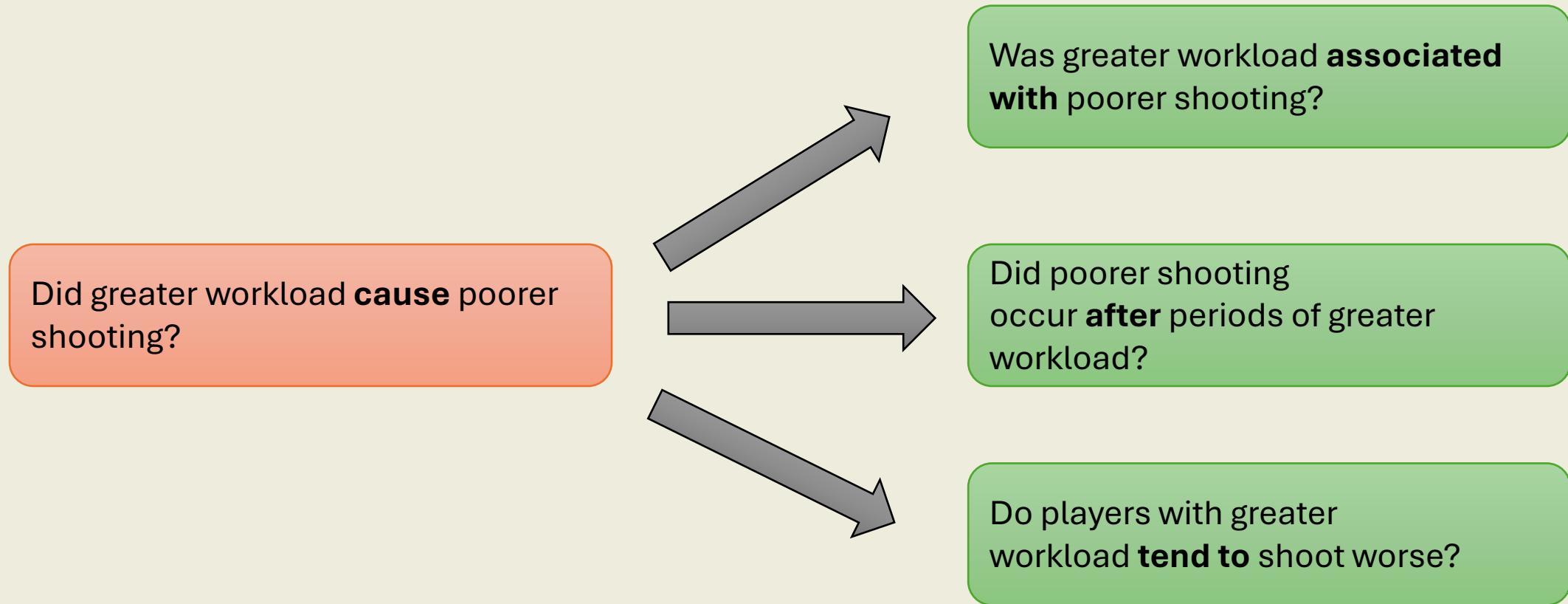


- What makes a basketball team successful?
- How does turnover rate relate to offensive efficiency across conference games?
- Did the turnover with 1:12 remaining in Q3 cause Tuesday's loss?

A useful scope is narrow enough to **investigate reasonably** and broad enough to **matter**.

Warning on Causal Language

The wording of the question should match what the evidence can support.



Activity: Diagnose the Question

“Does time spent training affect performance?”

With your group, identify what is missing or ambiguous:

- What decision or purpose?
- What outcome?
- What comparison?
- Which athletes and timeframe?
- What does “affect” claim?

Then rewrite the question.

EXAMPLE solution

Decision

- Should the coaching staff modify practice volume before games?

Question

- “Among players who appear in at least 15 minutes, is PlayerLoad accumulated during the two days before a game associated with fourth-quarter shooting efficiency?”

Still open for refinement

- Is PlayerLoad the most relevant measure?
- Is shooting efficiency the right outcome?
- What other context might explain the relationship?

The Streetlight Effect



Available Data Can Rewrite Questions

Original concern

- Why does the team struggle to create good shots against zone defense?

Available spreadsheet

- Points, rebounds, assists, and minutes

Convenient replacement question

- Who scores the fewest points?

The replacement is answerable, but it **does not answer** the original concern.

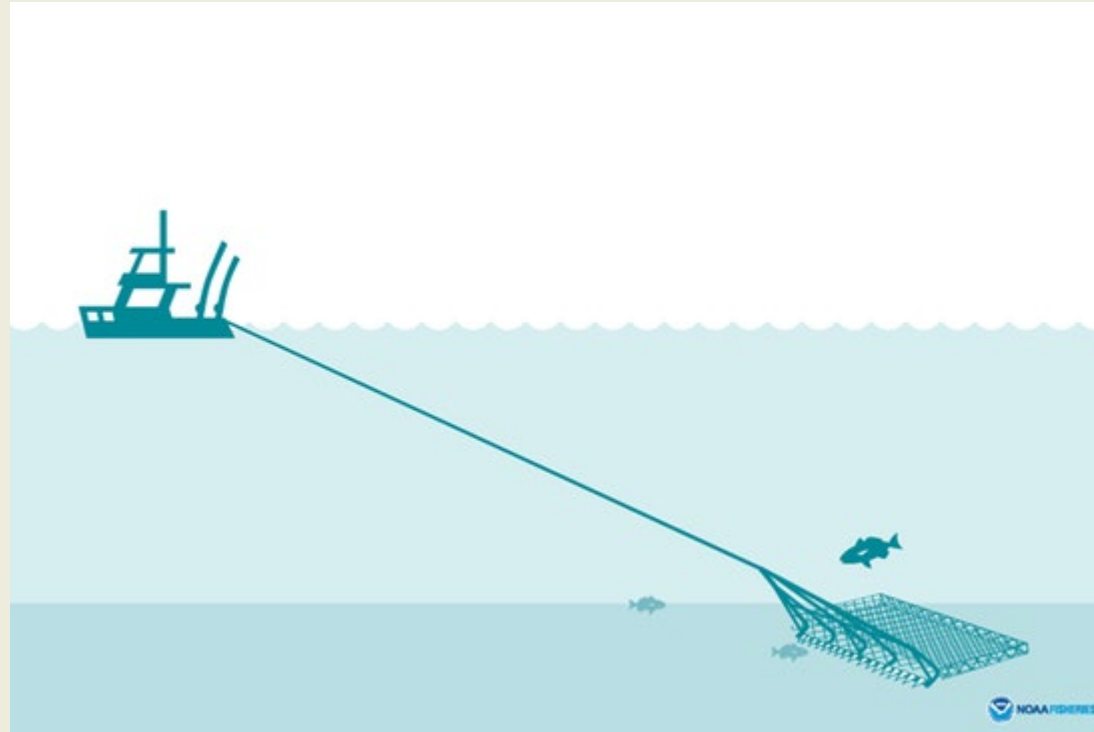
Data Availability Matters (But It Should Not Lead)

Practical Steps:

1. **Define** the sport **concern** and **decision**.
2. Write the **ideal question**.
3. **Identify the evidence** the question requires.
4. **Inspect** what data are actually available.
5. Revise the **scope** without changing the question's purpose.
6. State what the **available data cannot establish**.

Step 5 should refine the question, not replace it.

Data Dredging



probing data using unplanned analyses and then reporting salient results without accurately describing the processes by which the results were generated. (Erasmus, Holman & Ioannidis , 2022)

Data Dredging

- Most often done by repeated statistical testing
- Can produce misleading or false conclusions

“We have GPS data. What relationships can we find?”

- Statistically likely that you will find a significant relationship

Exploratory Analysis

- This is a legitimate analytical pursuit but stops at descriptive analytics.
- Valid diagnostic, predictive and prescriptive analytics requires direction from the start.

A Question Can Be Useful Even If the Answer Is “We Cannot Tell”

Sometimes the conclusion of an investigation may be:

- The required variable was not measured.
- The comparison group is inappropriate.
- The sample is too limited.
- The measurement does not represent the intended concept.
- The design cannot support the claimed explanation.

Recognizing limits is a crucial part of analytics

Group Project

Your group will develop a sport analytics project that connects:

- A sport observation, concern, or decision
- A focused analytical question
- Appropriate and obtainable data
- An analysis that matches the question
- A clear interpretation for the intended audience

Your project is not **“analyze a dataset”**. The dataset is **evidence** for a question.

CS1: Establish the Project Direction

With your group, identify:

1. A sport and domain that interests you
2. A real observation, concern, or decision
3. Who would use the answer
4. An initial analytical question
5. The ideal evidence needed
6. Data sources that may be feasible

Keep the concern (step 2) and the question (step 4) separate

Is the Question Ready to be Pursued?

Before committing, ask:

- Does the answer matter to someone?
- Is the outcome clearly defined?
- Is there a meaningful comparison?
- Is the scope realistic?
- Could evidence challenge our initial belief?
- Are we answering the concern—or merely using convenient data?

Summary & Looking Ahead

- Start with a sport concern or decision.
- Define the outcome, comparison, and context.
- Use causal language only when the evidence supports it.
- Let feasibility refine the scope without replacing the purpose.
- Treat limitations as part of the answer.
- Next: at-home case study (generate question and identify dataset)